

TRADEVault

Paper Trading Platform

Product Design & Technical Specification Document

React + Vite • TypeScript • TailwindCSS

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01 Project Overview & Vision

1.1 What is TradeVault?

TradeVault is a zero-risk, real-data paper trading web application built with React + Vite. It mirrors the full workflow of a professional brokerage platform — live market quotes, portfolio analytics, watchlists, order management, and P&L reporting — while executing all trades in a virtual, simulated environment. No real money is ever at risk.

The platform targets three audiences: beginners learning market mechanics, intermediate traders stress-testing new strategies, and educators needing a classroom-safe environment.

1.2 Core Platform Philosophy

Principle	Description	Implementation
Real Data, Fake Money	Live market prices via public APIs	Yahoo Finance / Polygon.io / Alpaca
Zero-Risk Learning	Virtual portfolio, no real execution	In-memory order book simulation
Professional UX	Same feel as Robinhood / Zerodha	Dark-mode first, chart-heavy UI
Performance	Sub-200 ms UI interactions	React Query + Vite code-splitting
Mobile-Ready	Responsive down to 375 px	TailwindCSS breakpoints

1.3 Tech Stack Summary

Layer	Technology	Purpose
Frontend Framework	React 18 + Vite 5	SPA with HMR dev server
Language	TypeScript 5	Type-safe codebase
Styling	TailwindCSS 3 + shadcn/ui	Utility-first + accessible components
Charts	TradingView Lightweight Charts	Candlestick, line, volume charts
State Management	Zustand + React Query	Global store + server cache
Routing	React Router v6	File-system-style routes
Market Data	Polygon.io WebSocket + REST	Real-time quotes & OHLCV
Auth	Supabase Auth (JWT)	Email/Google/GitHub OAuth
Database	Supabase PostgreSQL	Portfolio, orders, watchlists

Layer	Technology	Purpose
Deployment	Vercel	Edge CDN + preview deploys

02 Landing Page Design

2.1 Landing Page Goals

The landing page serves as the primary conversion surface. Its job is to communicate value within 3 seconds, establish trust, and route the visitor toward signup. The design follows the AIDA model (Attention → Interest → Desire → Action).

Design North Star

The landing page should feel like a fintech product that competes with Robinhood, Webull, and Zerodha — dark, data-rich, confident, and modern. First fold must load under 1.2 s on 4G.

2.2 Page Architecture (Section-by-Section)

2.2.1 Navigation Bar

Height	64 px fixed, backdrop-blur glass morphism effect
Left	TradeVault logo (SVG) + wordmark
Center	Markets • Features • Pricing • Blog (hidden on mobile)
Right	Log In (ghost button) + Start Free (accent filled button)
Scroll behavior	Transparent at top → solid dark background on scroll (transition 300 ms)
Mobile	Hamburger menu, full-screen drawer with same links

2.2.2 Hero Section

The most critical fold. Must communicate what the product is in one headline and one sub-headline, with a primary CTA above the fold on all screen sizes.

Element	Content / Spec
Background	Animated dark gradient (#0F172A → #1E1B4B) with subtle stock-ticker particle system (canvas)
Headline	"Master the Market. Risk Nothing." — 64 px, bold, white
Sub-headline	"Trade stocks with \$100,000 virtual cash. Real prices. Zero risk. Build your strategy before you go live." — 20 px, slate-400
Primary CTA	"Start Trading Free" — indigo-500 button, 52 px tall, hover scale-105

Element	Content / Spec
Secondary CTA	"Watch Demo" — ghost border button with play icon
Trust badge	"No credit card required • Free forever • Real market data"
Hero visual	Animated dashboard mockup (PNG/Lottie) floating right — shows chart + portfolio metrics
Stats strip	3 counters: 50,000+ traders • \$2.4B virtual volume • 98% uptime

2.2.3 How It Works

Three-step horizontal card flow. Animated icons. Simple language.

Step	Icon	Title	Body
1	Rocket	Sign Up Free	Create your account in 30 seconds. No personal finance data needed.
2	DollarSign	Get \$100K Virtual Cash	Your paper portfolio is loaded instantly with \$100,000 to invest.
3	TrendingUp	Trade Like a Pro	Place buy/sell orders, track P&L, and refine your strategy in real-time.

2.2.4 Features Section

Dark card grid (3 cols desktop, 1 col mobile). Each card has an icon, title, and 2-line description.

Feature Card	Icon	Description
Real-Time Charts	CandlestickChart	TradingView-powered charts with 50+ indicators
Live Market Quotes	Activity	Streaming prices via Polygon.io WebSocket
Order Types	ShoppingCart	Market, Limit, Stop-Loss, Trailing Stop orders
Portfolio Analytics	PieChart	Sector allocation, beta, Sharpe ratio
Watchlists	Bookmark	Custom watchlists with alerts
Trade Journal	BookOpen	Auto-log every trade with notes and tags

Feature Card	Icon	Description
Leaderboard	Trophy	Compete with other paper traders globally
Risk Simulator	Shield	What-if scenarios and VaR estimation

2.2.5 Social Proof Section

Layout	3-column testimonial card row
Content	Avatar + name + role + star rating + quote (max 100 chars)
Below cards	Logo strip: "As featured in" — TechCrunch, Product Hunt, IndieHackers

2.2.6 Pricing Section

TradeVault is freemium. Two tiers: Free and Pro.

Feature	Free	Pro (\$9/month)
Virtual portfolio	\$100,000	\$1,000,000
Watchlists	3	Unlimited
Chart indicators	10	50+
Trade history	30 days	Unlimited
Options paper trading	No	Yes
Crypto pairs	5	500+
AI strategy assistant	No	Yes
Export to CSV/PDF	No	Yes

2.2.7 Footer

Col 1	Logo + tagline + social icons (Twitter, GitHub, LinkedIn, Discord)
Col 2	Product: Features, Pricing, Changelog, Roadmap
Col 3	Company: About, Blog, Careers, Press
Col 4	Legal: Privacy Policy, Terms, Cookie Settings
Bottom bar	© 2026 TradeVault Inc. • Not financial advice. • Paper trading only.

03 Authentication Flow

3.1 Auth Pages

Authentication is handled by Supabase. All auth pages share a two-column layout: left side = brand visual / live market ticker, right side = form. This keeps the product visible during onboarding.

3.1.1 Sign-Up Page

Field / Element	Type	Validation
Full Name	text input	2–60 chars, letters/spaces only
Email	email input	RFC 5322 regex
Password	password input	Min 8 chars, 1 uppercase, 1 number — strength meter shown
Confirm Password	password input	Must match password field
Terms checkbox	checkbox	Required — links to ToS
Submit button	button	Disabled until all valid; shows spinner on submit
OAuth row	buttons	Continue with Google Continue with GitHub
Footer link	link	Already have an account? Log In

Post-Signup Flow

On success: Supabase sends verification email → user lands on /verify-email page → after click → redirected to /onboarding where they choose display name, avatar, and starting portfolio amount (\$10K, \$50K, or \$100K).

3.1.2 Login Page

Element	Behavior
Email + Password	Standard form, autofill-friendly
Remember Me	7-day session token via Supabase
Forgot Password	Link opens modal → sends reset email → /reset-password?token=...
OAuth	Google + GitHub (same as signup)
Failed attempts	3 failures → 30-second cooldown + CAPTCHA (hCaptcha)

3.1.3 Onboarding Wizard (3 steps)

1. Step 1 — Profile: Display name, avatar upload or avatar picker (8 presets), preferred currency (USD/INR/EUR/GBP).
2. Step 2 — Portfolio Setup: Starting balance selector (\$10K / \$50K / \$100K / Custom), trading style (Day Trader / Swing Trader / Investor), experience level (Beginner / Intermediate / Advanced).
3. Step 3 — Watchlist Seed: Pick 5+ symbols from popular list or search. These auto-populate the dashboard watchlist.

04 Post-Login Platform Overview

4.1 Application Shell

After login the user enters the main application shell — a persistent layout that wraps all authenticated pages. The shell consists of:

Shell Region	Width / Height	Content
Top Navbar	100% × 56 px	Logo, global search, market status pill, notifications bell, user avatar menu
Left Sidebar	240 px (collapsible to 64 px)	Navigation links — see Section 4.2
Main Content Area	Remaining width	Routed page content
Bottom Ticker Strip	100% × 32 px	Scrolling live prices for watchlist + major indices

4.2 Sidebar Navigation

Section	Links	Icon
Main	Dashboard, Markets, Screener	LayoutDashboard, Globe, Filter
Trading	Portfolio, Orders, Trade History, Watchlists	Briefcase, Receipt, Clock, Star
Analytics	P&L Report, Risk Analysis, Trade Journal	BarChart2, Shield, BookOpen
Social	Leaderboard, Community, Challenges	Trophy, Users, Zap
Account	Settings, Subscription, Help	Settings, CreditCard, HelpCircle

Sidebar Collapse

On screens < 1024 px sidebar auto-collapses to icon-only mode. On mobile (< 768 px) it hides fully and is accessible via the hamburger icon in the top navbar. Active route is highlighted with an indigo left-border + indigo text.

4.3 Global Top Navbar

Element	Description
Search (Cmd+K)	Global command palette — search symbols, pages, help docs. Fuzzy search with keyboard navigation.
Market Status Pill	Green dot + "Market Open" or Red dot + "Market Closed" with time-to-open/close
Notifications Bell	Badge with unread count. Dropdown shows: price alerts, order fills, system notices
Portfolio Value	Live \$ value of virtual portfolio — updates every 5 s during market hours
Avatar Menu	Dropdown: Profile, Subscription, Dark/Light Toggle, Log Out

4.4 Pages Overview

Route	Page Name	Purpose
/dashboard	Dashboard	Primary hub — charts, portfolio snapshot, news, watchlist
/markets	Markets	Browse all tradable symbols by sector/asset class
/screener	Screener	Filter stocks by technical/fundamental criteria
/portfolio	Portfolio	Holdings, allocation, open positions
/orders	Orders	Active, pending, and history of all orders
/trade-history	Trade History	Closed trades with P&L per trade
/watchlists	Watchlists	Manage multiple watchlists
/pnl	P&L Report	Realized + unrealized gains, charts
/risk	Risk Analysis	Beta, VaR, Sharpe, drawdown
/journal	Trade Journal	Annotated trade log with notes
/leaderboard	Leaderboard	Global + weekly rankings
/community	Community	Discussion threads per symbol
/settings	Settings	Profile, notifications, API keys, data export

05 Dashboard — Detailed Design

5.1 Dashboard Layout

The dashboard is a 12-column CSS Grid layout. It uses a "bento box" card system inspired by modern fintech dashboards (Robinhood, Webull, Groww). Every card is a draggable/resizable widget that saves layout to localStorage.

Design Reference

Primary inspiration: Robinhood Web (clean cards), Webull (data density), Trading 212 (color coding), and TradingView (chart quality). Dark theme is the default (#0F172A background, #1E293B card backgrounds).

5.2 Dashboard Cards / Widgets

5.2.1 Portfolio Summary Card (Top-Left, 4 cols wide)

Field	Display
Total Portfolio Value	Large \$XX,XXX.XX in white, +/- \$ and % change today
Buying Power	Available cash in green
Total Return	All-time return in \$ and %
Day's P&L	Today's gain/loss with color coding (green/red)
Mini sparkline	7-day portfolio value line chart

5.2.2 Main Chart Panel (Center, 8 cols wide, 2 rows tall)

This is the heart of the dashboard — a TradingView Lightweight Charts candlestick panel with full interactivity.

Feature	Specification
Chart type switcher	Candlestick Line Bar Area (tabs above chart)
Time frames	1m / 5m / 15m / 1h / 4h / 1D / 1W / 1M
Technical indicators	MA(20/50/200), EMA, VWAP, Bollinger Bands, RSI (below), MACD (below), Volume (histogram below main chart)
Symbol search bar	Within card — type to change symbol without leaving dashboard

Feature	Specification
Drawing tools	Trend lines, horizontal levels, Fibonacci retracement, text annotations
Full screen toggle	Expands chart to full viewport with ESC to exit
Data source badge	"Polygon.io • Real-time" shown in corner

5.2.3 Watchlist Card (Right sidebar, 4 cols)

Column	Content
Symbol	Ticker symbol bold
Price	Current price — flashes green on up-tick, red on down-tick
Change	+/- \$ and % for today
Sparkline	5-day mini line chart
Action	Quick Buy / Quick Sell buttons on row hover

- Search box at top to add symbols
- Tab bar to switch between saved watchlists
- Drag to reorder rows
- Right-click context menu: Set Alert, Add to Portfolio, Remove

5.2.4 Order Entry Panel (Right, below watchlist)

A compact order ticket always visible on the dashboard. No need to navigate away to place a paper trade.

Field	Options / Behavior
Symbol input	Auto-populated from clicked watchlist row or chart
Order side	Buy / Sell toggle (green/red background swap)
Order type	Market Limit Stop Stop-Limit Trailing Stop
Quantity	Shares input OR dollar amount toggle
Limit price	Shown only when Limit or Stop-Limit selected
Time in Force	Day GTC IOC FOK
Estimated cost	Auto-calculated: shares × price with commission = \$0
Buying power check	Inline warning if order exceeds available cash
Submit	"Place Buy Order" / "Place Sell Order" — confirm modal before submit

5.2.5 Portfolio Holdings Mini Table (Bottom-Left, 6 cols)

Column	Data
Symbol	Ticker + company name
Shares	Quantity held
Avg Cost	Average purchase price
Current Price	Live price
Market Value	Shares × current price
P&L	\$ and % unrealized gain/loss (color-coded)
Weight	% of total portfolio value

5.2.6 Market Overview Card (Bottom-Center, 6 cols)

A live snapshot of major market indices and sectors.

Panel	Content
Indices	S&P 500, NASDAQ, DOW, Russell 2000, VIX — price + % change
Sector heatmap	11 GICS sectors as color-coded squares (green = up, red = down)
Crypto strip	BTC, ETH, SOL, ADA prices with 24h change
Currency pairs	USD/INR, EUR/USD, GBP/USD, USD/JPY

5.2.7 Recent Trades Card (Bottom-Right, 4 cols)

Column	Data
Time	Execution timestamp
Symbol	Ticker
Side	BUY (green) / SELL (red)
Qty	Shares
Price	Execution price
P&L	For closed trades

5.2.8 News Feed Card (Optional, full-width bottom)

- Powered by Polygon.io /v2/reference/news API
- Filtered to watchlist symbols by default
- Each item: source logo + headline + symbol tags + relative timestamp
- Click opens article in side drawer (iframe + source attribution)

- Sentiment badge: Bullish / Bearish / Neutral (ML-classified)

06 Trading Functionality

6.1 Order Types Implementation

Order Type	Logic	Fill Condition
Market Order	Execute immediately at current bid/ask	Always fills at next available price tick
Limit Buy	Execute only at or below limit price	Fills when ask ≤ limit price
Limit Sell	Execute only at or above limit price	Fills when bid ≥ limit price
Stop Loss	Triggers market order when price hits stop	Converts to market when price ≤ stop
Stop-Limit	Triggers limit order at stop price	Converts to limit when stop triggered
Trailing Stop	Stop price trails by fixed \$ or %	Recalculates stop on each new high

6.2 Order Lifecycle State Machine

Every order passes through the following states:

State	Description	UI Color
PENDING	Order created, awaiting validation	Gray
OPEN	Resting in order book, waiting for fill condition	Amber
PARTIALLY_FILLED	Multi-lot orders, some shares filled	Blue
FILLED	Order fully executed	Green
CANCELLED	Cancelled by user or expiry (Day order after close)	Red
REJECTED	Insufficient buying power or invalid parameters	Dark Red

6.3 Portfolio Calculation Engine

All portfolio math runs client-side in a Zustand store with React Query for real-time price sync.

Metric	Formula
Position P&L	$(\text{Current Price} - \text{Avg Cost}) \times \text{Shares}$
Portfolio Return %	$(\text{Total Value} - \text{Starting Capital}) / \text{Starting Capital} \times 100$
Day P&L	Sum of $(\text{Current Price} - \text{Previous Close}) \times \text{Shares}$ for all holdings
Realized P&L	Sum of $(\text{Sell Price} - \text{Avg Cost}) \times \text{Shares}$ for each closed trade
Unrealized P&L	Sum of $(\text{Current Price} - \text{Avg Cost}) \times \text{Shares}$ for open positions
Sharpe Ratio	$(\text{Mean Daily Return} - \text{Risk-Free Rate}) / \text{StdDev Daily Returns} \times \sqrt{252}$
Max Drawdown	$\text{Max}(\text{Peak Value} - \text{Trough Value}) / \text{Peak Value}$ in rolling window
Win Rate	$\# \text{ Winning Trades} / \text{Total Closed Trades} \times 100$

6.4 Real-Time Data Architecture

Market data flows through a single WebSocket connection managed by a React Context provider.

Layer	Technology	Update Frequency
WebSocket connection	Polygon.io Stocks WebSocket	Per-tick during market hours
Quote normalization	Custom transformer hook	Every tick
UI subscription	Zustand subscribe()	Throttled to 1 update/500 ms per symbol
Stale data fallback	React Query refetch on reconnect	On WS disconnect
Pre-market / After-hours	Extended hours data flag	15-min delayed quotes
Historical OHLCV	Polygon.io REST API	On-demand per chart load

07 Additional Key Pages

7.1 Markets Page (/markets)

- Tabbed interface: Stocks | ETFs | Crypto | Forex | Futures
- Sortable data table: Symbol, Name, Price, Change%, Volume, Market Cap, P/E Ratio
- Sector filter chips (Technology, Healthcare, Finance, Energy, etc.)
- Quick-chart sparkline on row hover
- Click row → opens side drawer with Symbol Detail (mini chart + stats + news)

7.2 Stock Screener (/screener)

Advanced filter builder with saved screens. Inspired by Finviz and TradingView Screener.

Filter Category	Available Filters
Price	Price, Price Change %, 52W High/Low %
Volume	Volume, Relative Volume, Average Volume
Fundamentals	P/E, EPS, Revenue Growth, Dividend Yield, Market Cap
Technical	RSI, MACD Signal, Above/Below MA(20/50/200), ATR
Float & Short	Float, Short Interest %, Days to Cover

7.3 Portfolio Page (/portfolio)

Full portfolio analytics in one page:

- Donut chart: sector allocation + asset class split
- Holdings table (same columns as dashboard mini table, plus more detail)
- Performance chart: portfolio value vs S&P 500 benchmark over time
- Risk metrics card: Beta, Volatility, Sharpe, Max Drawdown, Correlation
- Dividend tracker (if any dividend-paying stocks held)

7.4 Trade Journal (/journal)

- Every paper trade auto-logged with timestamp, price, quantity, P&L
- User can add: Tags (Breakout, Earnings Play, Reversal, etc.), Notes (rich text), Screenshot (upload chart PNG), Star rating 1-5

- Filter by date, symbol, tag, rating
- Analytics panel: avg return by tag, win rate by strategy, trade frequency heat-map

7.5 Leaderboard (/leaderboard)

Tab	Ranking Criteria
All-Time	Highest total % portfolio return since account creation
This Week	Best % return in rolling 7-day window
This Month	Best % return in current calendar month
Challenges	Specific challenge scoreboards (e.g., "Earn 10% in 30 days")

Each leaderboard row shows: Rank, Avatar, Username, Return %, Portfolio Value, Win Rate, Badges.

08 Project File Structure

8.1 Vite + React Directory Layout

Convention
Feature-based folder structure. Each feature (auth, dashboard, trading, etc.) is self-contained with its own components, hooks, and types. Shared UI primitives live in src/components/ui.

Path	Purpose
src/main.tsx	App entry point, providers setup
src/App.tsx	Router tree, auth guard
src/components/ui/	shadcn/ui + custom primitives (Button, Card, Modal, etc.)
src/components/layout/	AppShell, Sidebar, Topbar, TickerStrip
src/features/auth/	Login, Signup, Onboarding pages + hooks
src/features/dashboard/	Dashboard page + all widget components
src/features/trading/	OrderEntry, OrderBook, TradeHistory components
src/features/markets/	MarketsTable, SymbolDrawer, Screener
src/features/portfolio/	Holdings, Allocation, Performance, Risk
src/features/charts/	ChartPanel, ChartToolbar, IndicatorManager
src/features/journal/	JournalTable, TradeNote, AnalyticsPanel
src/hooks/	useWebSocket, useQuotes, usePortfolio, useOrders
src/store/	Zustand stores: portfolioStore, ordersStore, watchlistStore
src/lib/	API clients, utils, calculations, constants
src/types/	TypeScript interfaces for Order, Position, Quote, etc.
src/styles/	globals.css, tailwind.config.ts

9.1 Phase 1 — Project Bootstrap (Week 1)

4. Scaffold: `npm create vite@latest tradevault -- --template react-ts`
5. Install dependencies: `tailwindcss`, `shadcn/ui`, `react-router-dom`, `zustand`, `@tanstack/react-query`, `lightweight-charts`, `lucide-react`
6. Configure Tailwind with custom colors and dark mode (class strategy)
7. Set up Supabase project — enable Auth (Email + Google), create tables: `profiles`, `portfolios`, `orders`, `positions`, `watchlists`, `journal_entries`
8. Create environment variables: `.env.local` with `VITE_SUPABASE_URL`, `VITE_SUPABASE_ANON_KEY`, `VITE_POLYGON_API_KEY`

9.2 Phase 2 — Landing Page (Week 2)

9. Build Navbar component with scroll-aware styling
10. Build HeroSection with animated gradient background and CTA buttons
11. Build HowItWorks 3-step cards
12. Build FeaturesGrid with lucide icons
13. Build PricingSection with two-tier comparison table
14. Build TestimonialsRow
15. Build Footer
16. Wire Signup CTA to `/signup` route

9.3 Phase 3 — Auth & Onboarding (Week 3)

17. Build SignupPage with Supabase `createUser`
18. Build LoginPage with session management
19. Build OnboardingWizard 3-step form
20. Build AuthGuard HOC to protect dashboard routes
21. Test email verification flow end-to-end

9.4 Phase 4 — App Shell (Week 4)

22. Build AppShell layout with Sidebar + Topbar + TickerStrip
23. Implement Sidebar collapse and mobile drawer
24. Build global CommandPalette (Cmd+K) with symbol search
25. Wire up React Router v6 nested routes within shell

9.5 Phase 5 — Dashboard & Charts (Weeks 5–6)

- 26. Integrate Polygon.io WebSocket for live quote streaming
- 27. Build ChartPanel using TradingView Lightweight Charts
- 28. Add chart type switcher and time-frame tabs
- 29. Add indicators panel (MA, EMA, RSI, MACD, Volume)
- 30. Build WatchlistCard with live price updates
- 31. Build OrderEntryPanel with all order types
- 32. Build PortfolioSummaryCard and MarketOverviewCard
- 33. Wire all cards to Zustand stores

9.6 Phase 6 — Trading Engine (Week 7)

- 34. Build order simulation engine (market fill logic)
- 35. Implement limit/stop order queue (check on every price tick)
- 36. Build portfolio calculation service (P&L, cost basis)
- 37. Build order lifecycle state machine
- 38. Add order confirmation modals with risk warning

9.7 Phase 7 — Analytics & Social (Weeks 8–9)

- 39. Build full Portfolio page with allocation donut chart
- 40. Build P&L Report page with recharts area chart
- 41. Build Risk Analysis page (Sharpe, drawdown, beta)
- 42. Build Trade Journal with CRUD operations
- 43. Build Leaderboard with real-time Supabase subscriptions

9.8 Phase 8 — Polish & Deploy (Week 10)

- 44. Responsive QA on mobile, tablet, desktop
- 45. Lighthouse audit: target > 90 Performance, 100 Accessibility
- 46. Add error boundaries and loading skeletons for all async components
- 47. Write E2E tests with Playwright for critical flows (signup → first trade → portfolio check)
- 48. Deploy to Vercel with environment variables
- 49. Set up Vercel preview deployments for PRs

10

Comparison with Current Market Apps

10.1 Competitive Landscape

App	Strengths	Weaknesses	TradeVault Advantage
Robinhood	Beautiful UX, simple order flow	No paper trading, US-only	Risk-free + global access
Webull	Paper trading, advanced charts	Cluttered UI, complex onboarding	Cleaner UI, faster onboarding
TradingView	Best charting platform	No portfolio management, paid	Full portfolio + charting, free tier
Investopedia Sim	Education-focused	Outdated UI, 15-min delay	Modern UI, near-real-time data
Thinkorswim	Institutional-grade tools	Steep learning curve, desktop only	Web-first, beginner-friendly
moomoo	Good paper trading, social	Complicated layout	Cleaner social + journal features

10.2 UX Patterns Adopted from Market Leaders

Pattern	Inspired By	Implementation in TradeVault
Bottom order ticket on mobile	Robinhood mobile	Fixed bottom drawer on < 768 px
Green/red flash on price tick	Bloomberg Terminal	CSS keyframe flash animation
Bento box dashboard	Groww, Smallcase	CSS Grid draggable widget system
Candlestick chart quality	TradingView	Lightweight Charts library
Dark theme default	Webull, Binance	Tailwind dark class strategy
Cmd+K command palette	Linear, Vercel	cmdk library integration
Real-time leaderboard	eToro	Supabase Realtime subscriptions
Trade journal with tags	Tradervue, Edgewonk	Custom CRUD with rich text
Sector heatmap	Finviz	D3-powered SVG treemap
News sentiment badges	Bloomberg, Seeking Alpha	Claude API for NLP classification

10.3 Regulatory & Compliance Notes

Important Disclaimer

TradeVault is a paper trading simulation platform. All trades are virtual. No real securities transactions occur. The platform must prominently display: "This is not financial advice. All trading is simulated. Past paper trading performance does not predict real market results."

- Display disclaimer in footer, on every order confirmation, and in onboarding
- No real money, no payment processing required for free tier
- Pro subscription via Stripe (no financial instrument handling)
- GDPR-compliant data deletion on account closure (Supabase RLS policies)
- Market data re-distribution: Polygon.io license allows display in apps (verify per plan)

A **Appendix — Key Dependencies**

Package	Version	Purpose	License
react	^18.3	UI framework	MIT
vite	^5.4	Build tool + dev server	MIT
typescript	^5.5	Type safety	Apache 2.0
tailwindcss	^3.4	Utility CSS	MIT
@shadcn/ui	latest	Accessible UI components	MIT
react-router-dom	^6.26	Client-side routing	MIT
@tanstack/react-query	^5.56	Server state management	MIT
zustand	^4.5	Global client state	MIT
lightweight-charts	^4.2	TradingView charts	Apache 2.0
@supabase/supabase-js	^2.45	Auth + database	MIT
lucide-react	^0.451	Icon library	ISC
recharts	^2.12	Analytics charts	MIT
cmdk	^1.0	Command palette	MIT
date-fns	^3.6	Date utilities	MIT
zod	^3.23	Schema validation	MIT

End of TradeVault Product Design & Technical Specification Document v1.0

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